# Does a JEPA world model help fly an unidentified fixed-wing aircraft?

A controlled test of the SkyJEPA recipe on a simulated Ultra Stick 120

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**Answer in one line**

**In this setting, no.** On a fixed wing whose commands are not forces, **JEPA + physics** reproduces SkyJEPA's ranking on training-range aircraft, but the two controls SkyJEPA did not run predict better (the same **physics head** trained end to end, and a **Plain MLP** of 10.6k parameters), and the **JEPA** latent keeps little motor or CG information.

## 1 The problem

**Goal.** Fly a fixed-wing aircraft whose aerodynamics, motor and center of gravity are unknown (new airframe, new payload or propeller) from its last measurements, with noisy sensors, without an identification flight.

**What exists**

- **System identification**: one dedicated test campaign per airframe, redone after every change.
- **Adaptive control** (L1 [9], INDI [10]): adapts the controller but gives no multi-step model to plan with.
- **Learned adaptation**: RMA [6] (policy trained with the true parameters), Neural-Fly [11] (learned residual forces with online adaptation).

**Approach studied.** A learned **world model inside MPPI** [4], which predicts how the current aircraft will move.

## 2 What SkyJEPA claims, and what it did not test

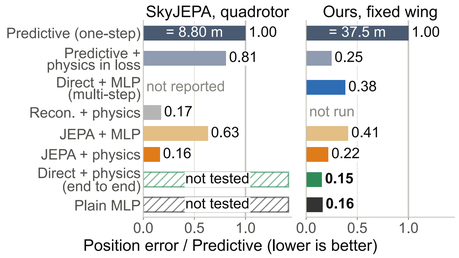
SkyJEPA [1]: a JEPA world model [3] of a quadrotor (TCN encoders, GRU predictor, SIGReg [2]) plus a physics-informed prober, trained in simulation on randomized vehicles, used in MPPI.

- Their Table III (horizon not stated): **JEPA + physics prober 1.43 m**, best; reconstruction + **physics prober 1.53 m**; **JEPA** + MLP prober **5.56 m**; Predictive **8.80 m**.
- With the **physics prober**, JEPA adds only **0.10 m** over reconstruction (1.43 vs 1.53 m).

Their actions are the four motor forces, so the thrust is known from the action:

$$\dot{v} = \frac{\sum_i f_i}{m} R\, e_3 - g + \Delta\dot{v}$$

**Not tested**: a plain MLP, and the physics prober on a predictive (state-loss) model, frozen or trained end to end.

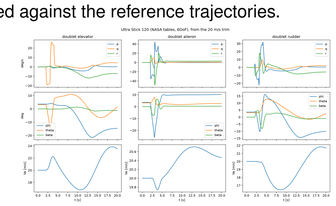



Position error divided by the one-step Predictive model (SkyJEPA: 8.80 m; ours: 37.5 m at 5 s, training-range aircraft, mean of 3 seeds).

**Q1** Does JEPA predict better?
**Q2** Does it identify the aircraft without labels?
**Q3** Is it less sensitive to sensor noise at equal training?
**Q4** Does it fly better in closed loop?

## 3 Test bench

**Simulator.** Python port of the University of Minnesota Ultra Stick 120 6-DoF model [8] (NASA wind-tunnel tables), checked against the reference trajectories.



**Data. 2,000** training flights of 30 s in calm air (steady wind, no turbulence or gusts) and **200** test flights of 120 s at 20 Hz, flown by a TECS autopilot [12] with excitation maneuvers. Every flight has its own aircraft:

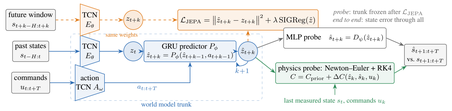
| randomized | training | extreme* |
|---|---|---|
| mass | ±20 % | 20–35 % |
| inertias | ±30 % | 30–45 % |
| aero coefficients (6, each) | ±20 % | 20–30 % |
| motor power | ±30 % | 30–45 % |
| CG shift (of chord) | ±3 % | 3–5 % |

* half the test flights, all parameters at once

**Commands** are elevator, aileron, rudder, throttle: **not forces**. This is the key difference with SkyJEPA.

**State**: 9 quantities (airspeed, angle of attack, sideslip, $p$, $q$, $r$, roll, pitch, vertical speed).

**Sizes as in SkyJEPA**: state TCN [8,8,16], action TCN [4,4,8], GRU 24, $z \in \mathbb{R}^{24}$, SIGReg $\lambda = 0.02$, training horizon 20 steps; history 11 steps = 0.55 s (SkyJEPA: 10); the estimated wind is also an input.



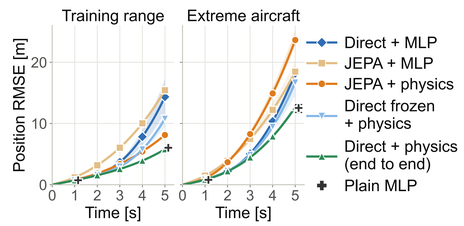

probe = head trained on the frozen trunk.

**Models compared**, 10.6–11.5k parameters (MLP 10,559; latent models 10,713; physics-head models 11,542):

| this work | SkyJEPA name |
|---|---|
| **Predictive (one-step)** | Predictive |
| **Direct + MLP** (state loss over 1-s rollouts) | not reported |
| **JEPA + MLP** | JEPA + prober |
| **JEPA + physics** | JEPA + PI prober (their recipe) |
| **Direct frozen + physics** | not tested |
| **Direct + physics (end to end)** | not tested |
| **Plain MLP**: raw 0.55-s window, next state = last state + MLP [7] | not tested |

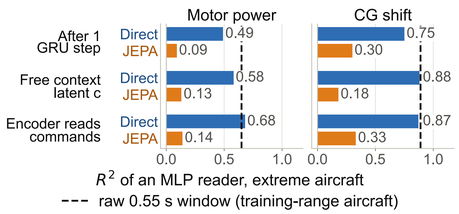
**Physics head**: outputs force and moment coefficients, is given the true mass and inertias (no aerodynamic prior), and integrates Newton–Euler with RK4 from the last measured state. The MLP-head models and the **Plain MLP** get no aircraft parameters.

## 4 Q1, open loop: SkyJEPA's ranking holds, the missing controls win




- Training-range aircraft, 5 s, 3 seeds: we reproduce SkyJEPA's ranking. **JEPA + physics 8.1 m** beats **JEPA + MLP 15.4 m**, **Direct frozen + physics 10.8 m** and **Predictive (one-step) 37.5 m**.
- The two controls SkyJEPA did not run do better: **Direct + physics (end to end) 5.8 m**, **Plain MLP 6.0 m**.
- Extreme aircraft: **JEPA + physics** is the worst, **23.6 m** (**JEPA + MLP** 18.5, **Plain MLP** 12.5, **Direct + physics (end to end)** 12.7).

## 5 Q2, identification: the JEPA latent drops the aircraft




$R^2$ of an MLP reader per aircraft parameter (learned representations on extreme aircraft, raw window on training-range aircraft).

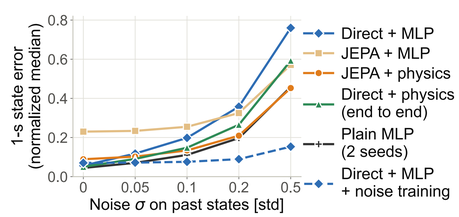
- The raw 0.55-s window (states + commands) holds the motor (**0.65**) and the CG (**0.89**).
- **Direct + MLP** recovers them after one GRU step (**0.49** / **0.75**).
- **JEPA** does not (**0.09** / **0.30**), even when its encoder also reads the commands (**0.14** / **0.33**, before the GRU), or with a free context latent $c$ that is not a prediction target (**0.13** / **0.18** vs **0.58** / **0.88** for the **Direct** trunk with $c$).
- Mass and $C_L$ are read by both (about 0.5).

**Why.** In a JEPA the GRU memory is the latent itself, and the loss pins $\hat{z}_{t+k}$ to the encoding of a state window, which holds almost no motor or CG information ($R^2 \approx 0$). Why the free latent stays poor is open; one hypothesis: the objective favors slow, easy-to-predict content.

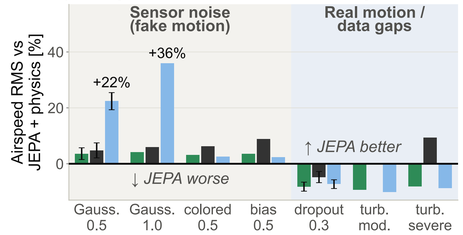
**Consequence.** A frozen **Direct** trunk + **physics head** matched end-to-end training only with a free context latent (training range / extreme):

| | |
|---|---|
| **Direct trunk + context, frozen, + physics**, seed 0 | **6.0** / **11.1 m** |
| **Direct + physics (end to end)**, seed 0 | **5.9** / **12.2 m** |
| **Direct frozen + physics**, no context, 3 seeds | **10.8** / **16.7 m** |
| **JEPA trunk + context, frozen, + physics**, seed 0 | **8.0** / **20.6 m** |

## 6 Q3, noise at equal (clean) training: JEPA acts like a low-pass filter (interpretation)




**Open loop** (training-range aircraft, mean of seeds): **JEPA + MLP** starts high (**0.23** vs **0.045** for the **Plain MLP** at $\sigma = 0$) and grows less in relative terms ($\times 2.5$ vs $\times 10$ at $\sigma = 0.5$); it stays above the **Plain MLP** at every $\sigma$.




Trained on clean data, one training seed per model; 200 aircraft; env. seed 1. Error bars: paired bootstrap 95% CI.

Direct + physics (end to end) · Plain MLP · Direct frozen + physics

**Closed loop** (MPPI, 200 aircraft, environment seed never used for tuning, models trained on clean data, one training seed each; tracking over the aircraft every controller kept):

- **JEPA + physics** tracks airspeed **3–9 %** better than **Direct + physics (end to end)** and the **Plain MLP** under the four sensor-noise kinds. Gaussian 0.5: **1.19** vs **1.24** and **1.25 m/s** (paired bootstrap 95 % CI excludes 0, 177 common aircraft); Gaussian 1.0: **1.67** vs **1.74** and **1.77**.
- Altitude tracking is mixed (**Plain MLP** better under Gaussian 0.5: 7.69 vs 7.91 m).
- Worse than **Direct + physics (end to end)** under real motion and data gaps: severe turbulence **1.49** vs **1.37 m/s** (**Plain MLP** 1.63); dropout 0.3: **1.05** vs **0.97**.

Smoothing would help when the fast content is fake motion and hurt when it is real. This is consistent with JEPA favoring predictable, low-noise features [5], but we did not test it directly.

**In progress: equal *noise* training.** A denoising **JEPA** vs **Direct + physics (end to end)** and the **Plain MLP**, trained with the same noise, tested on noise kinds never seen in training. Already measured: the **Plain MLP** trained with noise tracks at **1.20 m/s** under Gaussian 1.0 (**JEPA + physics**, clean: **1.67**), but 1.62 vs 1.56 m/s for the clean **Plain MLP** under severe turbulence (separate run, its own turbulence draw).

## 7 Q4, closed loop: the Plain MLP and every physics-head model fly

| Survivors out of 200 | Gauss. 0.5 | drop. 0.3 | Gauss. 1.0 | col. 0.5 | bias 0.5 | turb. mod. | turb. sev. |
|---|---|---|---|---|---|---|---|
| TECS autopilot | 200 | 200 | 200 | 200 | 200 | 200 | 200 |
| Plain MLP | 200 | 200 | 200 | 200 | 200 | 200 | 200 |
| Direct + MLP | 177 | 197 | – | – | – | – | – |
| Direct + physics (end to end) | 200 | 200 | 200 | 200 | 200 | 200 | 200 |
| JEPA + physics | 200 | 200 | 200 | 200 | 200 | 200 | 200 |
| JEPA + physics, context c | – | – | 200 | 200 | 199 | 200 | 200 |
| Direct frozen + physics | 200 | 200 | 200 | 200 | 200 | 200 | 200 |

Survivors out of 200; – = not run. Gauss. = Gaussian, col. = colored, drop. = dropout, turb. = turbulence.

- **TECS**, the **Plain MLP** and every **physics-head** model keep **199–200 of 200** aircraft in every condition they flew (7 conditions; the context variants 5).
- **Direct + MLP** (flown in 2 conditions) loses aircraft: **177**/**200** under Gaussian 0.5 (McNemar $p < 0.001$), **197**/**200** under dropout. **JEPA + MLP** was not flown here; it crashed every aircraft in an earlier 40-aircraft test.
- With the best model (**JEPA + physics**), MPPI tracks airspeed **32 %** better than **TECS** under Gaussian 0.5 (**1.19** vs **1.75 m/s**); **TECS** has no sensor filter and tracks roll better.

Survival does not separate the **Plain MLP** and the physics-head models; tracking does.

## 8 Conclusions

1. SkyJEPA's ranking reproduces on training-range aircraft, but the **physics head** trained end to end (**5.8 m**) and a **Plain MLP** (**6.0 m**) beat **JEPA + physics** (**8.1 m**), and **JEPA + physics** is the worst on extreme aircraft.
2. Here the commands are not forces, so the model must identify the airframe; the **JEPA** latent keeps little motor and CG information ($R^2$ **0.09** / **0.30** vs **0.49** / **0.75**), even with a free context latent.
3. At clean training, the **JEPA** trunk tracks **3–9 %** better under sensor noise and worse than the end-to-end physics model under turbulence: it smooths.
4. With 9 states, a **Plain MLP** of 10.6k parameters on the raw window predicts as well as the best physics model and better than the JEPA models.

**Where JEPA could still help**: rich observations with distractor channels or images (next step).

**Limits**: simulation only; one airframe family; one training seed per model in closed loop; physics heads receive the true mass and inertia, the other models do not; the link to "commands are not forces" is not tested by varying the command type.

**References**

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Code and results: `https://github.com/clementdumeril/WingJEPA`